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CCIT

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Protecting the Digital World, One Concept at a Time

Today's Agenda

- 1 Welcome & Course Overview
- 2 Learning Objectives
- 3 Course Structure & Schedule
- 4 Assessment & Grading
- 5 Introduction to Cybersecurity
- 6 Cybersecurity Domains
- 7 Expectations & Success Tips
- 8 Let's Get Started!
- 9 Looking Ahead

Welcome to Cybersecurity! ■

Why This Course Matters:

- Digital threats are **everywhere**
- Organizations lose \$billions annually
- Skills gap: 3.5M unfilled cybersecurity jobs
- **YOU** will be part of the solution



"The only system which is truly secure is one which is switched off and unplugged, locked in a titanium lined safe, buried in a concrete bunker, and is surrounded by nerve gas and very highly paid armed guards. Even then, I wouldn't stake my life on it." — Gene Spafford

Your Instructor & Course Details

Course Code: CCY2001

Credit Hours: 3

Prerequisites: CCS201

Contact Hours:

- Lectures: 2 hours/week
- Tutorials: 2 hours/week
- Lab: 0 hours (Self-study)

Key Resources:

- ○ Office Hours: [TBA]
- □ Email: [TBA]
- ► Textbook: "Cybersecurity Fundamentals: A Real-World Perspective" by Kutub Thakur & Al-Sakib Khan Pathan, CRC Press

Important

👉 Check the course portal regularly for updates, assignments, and announcements!

What You'll Learn This Semester

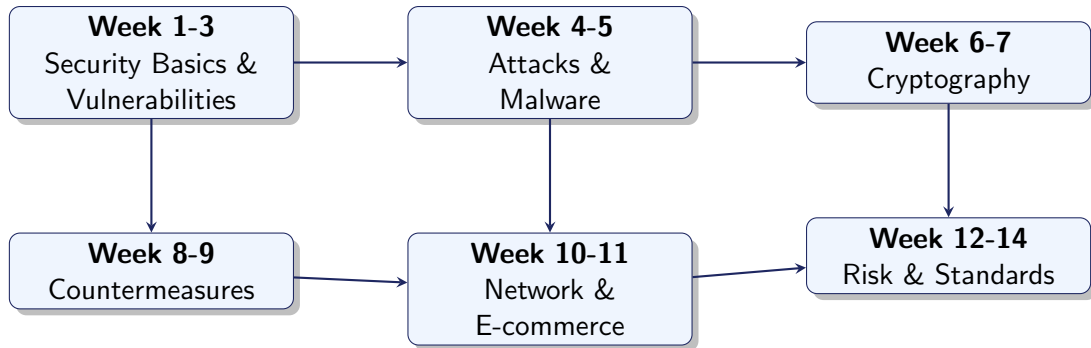
Core Competencies:

- 1 **Identify** security vulnerabilities
- 2 **Recognize** cyberattack types
- 3 **Understand** countermeasures
- 4 **Apply** cryptography basics
- 5 **Implement** network security
- 6 **Secure** E-commerce systems
- 7 **Manage** cybersecurity risks
- 8 **Differentiate** security standards

Learning Outcome

By the end of this course, you'll think like a cybersecurity professional — identifying threats before they become breaches.

Skills Map: From Beginner to Defender



15-Week Journey: Your Roadmap

Week	Topic
1	Course Introduction
2	Cybersecurity Basics
3	Cyberattacks and Vulnerabilities
4	Recent Cyberattacks and Their Impact
5	Computer Malware and Classification
6	Cryptography Basics – Part I
7	Cryptography Basics – Part II
8	Countermeasures for Cyberattacks – Part I

15-Week Journey (Continued)

Week	Topic
9	Countermeasures for Cyberattacks – Part II
10	Network Security Basics
11	E-commerce Security Basics
12	Cybersecurity Risk Management Basics
13	Cybersecurity Standards
14	Cybersecurity for the Future
15	Revision

How You'll Be Evaluated

Component	Marks	%
Quizzes (Weeks 1-7)	30	30%
Reports (Weeks 8-12)	20	20%
Presentations (Weeks 1-15)	10	10%
Final Exam (Week 16/17)	40	40%
Total	100	100%

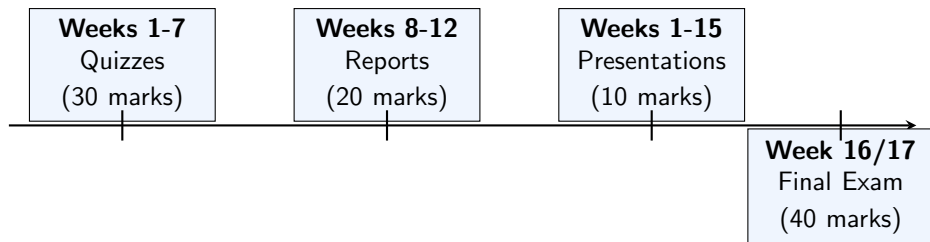
Visual Breakdown:

- Quizzes: 30%
- Reports: 20%
- Presentations: 10%
- Final Exam: 40%

Pro Tip

★ Continuous assessment = 60% of your grade. Stay engaged throughout the semester!

Assessment Timeline



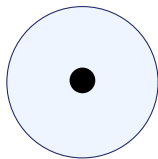
Key Dates:

- Quizzes: Every week (Weeks 1-7)
- Reports: Ongoing (Weeks 8-12)
- Final Exam: Week 16 or 17 (Check schedule)

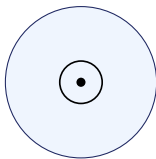
What IS Cybersecurity?

Definition

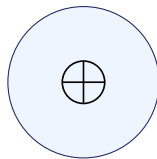
Cybersecurity is the practice of protecting systems, networks, programs, and data from digital attacks, unauthorized access, damage, or theft.



Protect



Detect



Respond

The CIA Triad: Foundation of Security

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Confidentiality

Only authorized access



Integrity

Data accuracy & trustworthiness



Availability

Accessible when needed

Remember

All security controls aim to protect one or more of these three principles!

Why Cybersecurity Matters: Real Stats

The Threat Landscape:

- 43% of cyberattacks target small businesses
- 95% of breaches due to human error
- Average breach cost: **\$4.45M** (2023)
- Attack every **39 seconds**

Most Common Threats:

- 1 × Malware & Ransomware
- 2 ~ Phishing Attacks
- 3 ♦ Social Engineering
- 4 ○ Cloud Vulnerabilities
- 5 ≡ DDoS Attacks

Wake-Up Call

By 2025, cybercrime damages will cost the world \$10.5 trillion annually!

The Good News

Demand for cybersecurity professionals continues to grow exponentially!

Real-World Examples: Major Breaches

Company	Year	Records Affected
Yahoo	2013-2014	3 billion accounts
Equifax	2017	147 million
Marriott	2018	500 million
Facebook	2019	533 million
LinkedIn	2021	700 million
Twitter	2022	5.4 million

Discussion Point ...

Think-Pair-Share: What do all these breaches have in common? What could have prevented them?
(2 minutes: *think individually, then discuss with your neighbor*)

The Cybersecurity Landscape

Key Domains You'll Explore:

① ≡ Network Security

- Firewalls, IDS/IPS
- VPNs, Network monitoring

② • Cryptography

- Encryption algorithms
- Digital signatures, PKI

③ ≡ Application Security

- Secure coding
- Web application security

④ ▷ E-commerce Security

- Payment systems
- SSL/TLS protocols

⑤ ↗ Risk Management

- Risk assessment
- Compliance frameworks

⑥ □ Standards & Policies

- ISO 27001, NIST
- Best practices

Career Paths in Cybersecurity

Security Analyst

Monitor &
respond

Penetration Tester

Ethical hacking

Security Architect

Design systems

Incident Re- sponder

Handle breaches

Cryptographer

Develop en-
cryption

CISO

Lead security

Forensics Expert

Investigate crimes

Compliance Officer

Ensure standards

Average Salaries (2024): \$80K - \$180K+ depending on role and experience

What We Expect From You

Academic Expectations:

- ✓ Attend all lectures & tutorials
- ✓ Complete readings before class
- ✓ Participate in discussions
- ✓ Submit work on time
- ✓ Practice ethical behavior

Professional Conduct:

- ∞ Respect classmates
- ⊘ No plagiarism/cheating
- ∞ Follow ethical guidelines
- ■ Use skills responsibly
- 📢 Report concerns

Ethical Hacking vs. Criminal Hacking

Skills learned in this class are for **defense only**. Unauthorized access to systems is **illegal** and will result in serious consequences!

Tips for Success

Study Strategies:

- 1 ► Review lectures regularly
- 2 ≡ Practice hands-on labs
- 3 ♣ Form study groups
- 4 □ Follow security news
- 5 ? Ask questions early

Resources to Explore:

- Krebs on Security (blog)
- SANS Reading Room
- OWASP Top 10
- Cybrary (free courses)
- HackTheBox (practice)

Growth Mindset

★ Cybersecurity is constantly evolving. Embrace continuous learning!

Academic Integrity Matters

Our Policy

Academic honesty is **non-negotiable**. All work must be your own unless explicitly stated as group work.

Prohibited Activities:

- × Copying code/answers from classmates or internet
- × Using unauthorized tools during exams
- × Submitting AI-generated work as your own
- × Sharing solutions before deadlines

Allowed Activities:

- ✓ Discussing concepts with classmates
- ✓ Using official course resources
- ✓ Seeking clarification from instructor
- ✓ Citing sources properly

Violations will be reported and may result in course failure.

Quick Poll: Your Background |

Question 1

How would you rate your current knowledge of cybersecurity?

- ① Beginner (Never studied it)
- ② Novice (Some awareness)
- ③ Intermediate (Taken related courses)
- ④ Advanced (Working experience)

Question 2

What interests you most about cybersecurity?

- ① Protecting systems from attacks
- ② Understanding how hackers work
- ③ Cryptography and encryption
- ④ Career opportunities

Challenge: Spot the Phishing Email ☐

Scenario

You receive this email. Is it legitimate or a phishing attempt?

Email Content

From: security-team@yourbank-verify.com

Subject: URGENT: Account Suspended - Verify Now

Dear Valued Customer,

Your account has been suspended due to suspicious activity. To restore access, click the link below within 24 hours:

<http://yourbank-verify.com/restore>

Failure to verify will result in permanent account closure.

Best regards,

YourBank Security Team

Discussion: What are the red flags? (2 minutes)

Phishing Red Flags Revealed

Warning Signs:

① Suspicious sender domain

- yourbank-verify.com
- Not official bank domain

② Urgency tactics

- “24 hours”
- “Permanent closure”

③ Generic greeting

- “Dear Valued Customer”
- No personal name

④ Suspicious link

- Doesn't match bank URL
- Asks for verification

⑤ Threatening tone

- Creates panic
- Demands immediate action

Best Practice

✓ Always verify directly with your bank through official channels, never through email links!

Next Week: Cybersecurity Basics

Week 2 Preview:

- Fundamental security concepts
- Threat actors and their motivations
- Attack vectors and surfaces
- Basic defense mechanisms
- Introduction to security tools

Preparation

Before Next Class:

- ① Read Chapter 1 of the textbook
- ② Create an account on the course portal
- ③ Review the syllabus thoroughly
- ④ Prepare questions for tutorial session

Where to Get Help:

- △ Office hours (Weekly)
- □ Email (48hr response)
- ... Discussion forum
- ♣ Study groups
- ► Course portal

Important Links:

- Course Portal: [TBA]
- Textbook Resources: [TBA]
- Lab Environment: [TBA]

Remember

! Don't wait until the last minute to ask for help!

Community

∞ We're all in this together. Support each other's learning!

Stay Updated

⦿ Enable notifications for course announcements



Your Questions?

Now is the time to ask anything about:

- Course structure
- Assessment methods
- Content coverage
- Resources
- Prerequisites
- Expectations
- Career paths
- Anything else!

“There are no silly questions, only the questions you don’t ask!”



Welcome to CCY2001!

Let's Build a Secure Digital Future Together

See you next week!

Stay curious, stay secure

Email: [TBA] — Office: [TBA] — Portal: [TBA]